Task 5: Ensemble Learning, Model Optimization & Real-World ML Pipeline

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1. Model Comparison

A classification pipeline was built on the Breast Cancer Wisconsin dataset (569 samples, 30 features; Malignant vs Benign) using a stratified 80/20 split. Four models were trained and evaluated on the same held-out test set: a Logistic Regression baseline (on scaled features), a Random Forest (bagging ensemble), a Gradient Boosting classifier (boosting ensemble), and a Random Forest tuned with GridSearchCV. Random Forest reduces overfitting by averaging many decorrelated trees, while Gradient Boosting builds trees sequentially so each one corrects the previous errors.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV F1 (5-fold)
Logistic Regression	0.9825	0.9861	0.9861	0.9861	0.9954	0.985 (±0.005)
Random Forest	0.9561	0.9589	0.9722	0.9655	0.9937	0.965 (±0.018)
Gradient Boosting	0.9561	0.9467	0.9861	0.9660	0.9907	0.971 (±0.017)
Random Forest (tuned)	0.9561	0.9589	0.9722	0.9655	0.9934	0.968 (±0.020)

(Logistic Regression row highlighted: best on every metric.)

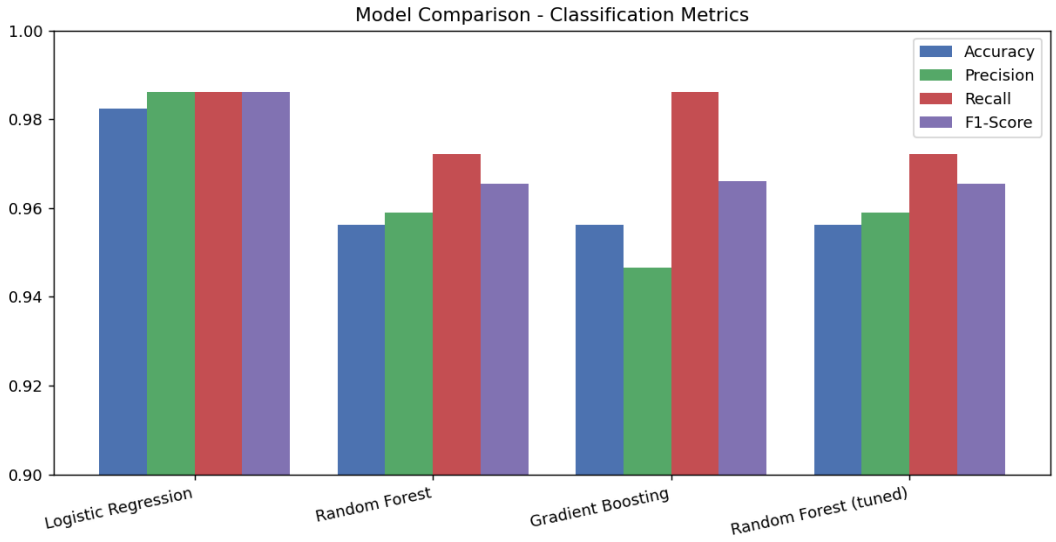


Figure 1: Accuracy, precision, recall and F1 for all four models.

2. Optimization Explanation (GridSearchCV)

GridSearchCV (3-fold) searched 6 combinations of n_estimators [50, 100] and max_depth [3, 5, 7] for the Random Forest. The best parameters were **max_depth=5, n_estimators=100** with a mean CV accuracy of 0.9538, versus about 0.949 for the weaker settings. The gain from tuning was small, and test-set accuracy stayed at 0.9561: the default forest was already near the ceiling for this model family on this data. This is a useful result in itself, since tuning should be judged by evidence and not assumed to help.

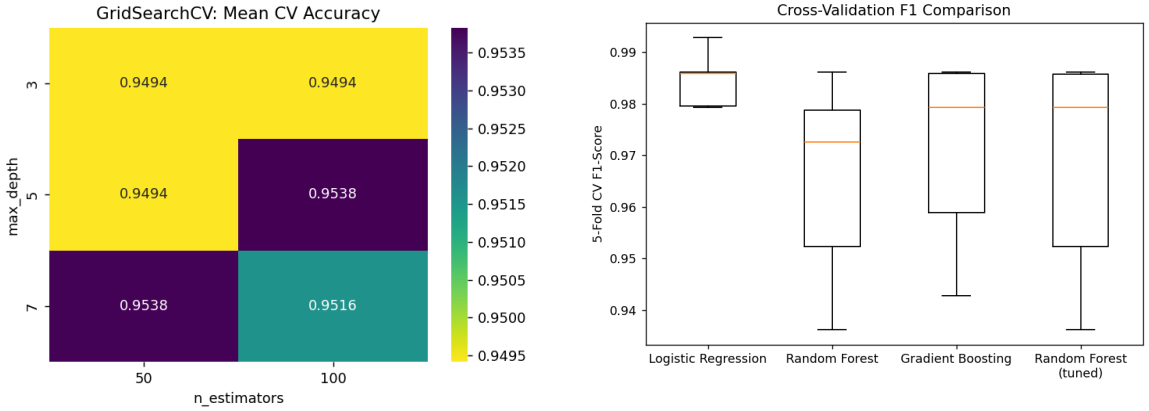


Figure 2 and 3: GridSearchCV accuracy grid (left) and 5-fold cross-validated F1 spread per model (right).

Cross-validation and pipeline: A single test split can be lucky or unlucky, so every model was also scored with 5-fold CV. Scikit-learn's Pipeline API was used for the Logistic Regression so scaling is fit only on training folds (no data leakage).

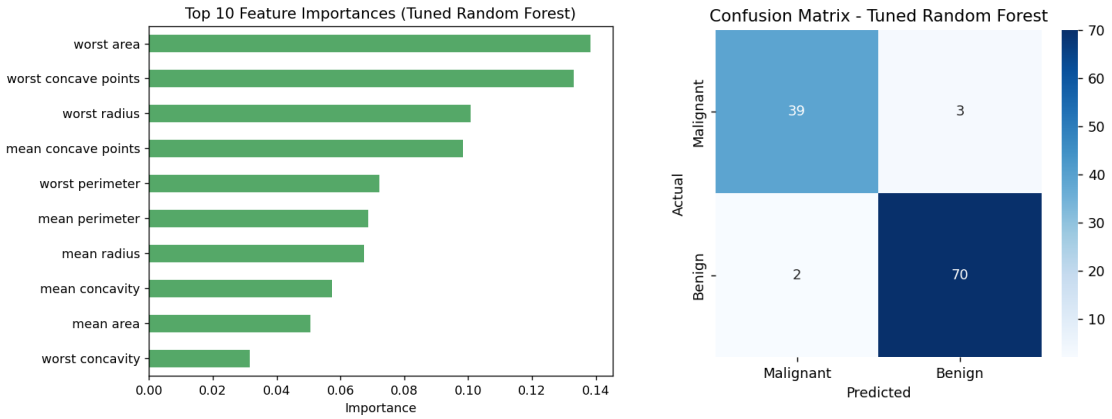


Figure 4 and 5: Top feature importances from the tuned Random Forest (left) and the final model's confusion matrix (right).

3. Final Decision

Selected model: Logistic Regression. It was chosen on the three criteria in the task brief:
• **Performance:** highest accuracy (0.9825), precision, recall, F1 (0.9861) and ROC-AUC (0.9954) on the test set. Its confusion matrix has only 1 false positive and 1 false negative out of 114 samples.
• **Stability:** best 5-fold CV F1 (0.985) with the smallest spread (±0.005); the ensembles varied roughly 3 to 4 times more between folds.
• **Interpretability:** its coefficients can be explained directly, which matters for medical decisions and for building clinician trust.

Why the ensembles did not win: the dataset is small and nearly linearly separable, so a linear model is already close to optimal, while Random Forest and Gradient Boosting add variance without extra signal to capture. Gradient Boosting matched the baseline's recall (0.986) but with lower precision, meaning more false alarms. Ensembles are expected to pay off on larger, noisier, non-linear problems such as fraud detection or recommendations; the professional takeaway is to benchmark them against a strong simple baseline before choosing complexity.

4. Deliverables

AI_ML_Task5_Ensemble_Learning.ipynb (model comparison, GridSearchCV output, final metrics, feature importance, CV comparison, Pipeline API), this PDF report, comparison CSV files, and the saved models (final_logistic_model.joblib, best_rf_model.joblib, scaler.joblib).